

LAURENT "ENZO" NTIBARIKURE

Engineering | Energy & Information Systems | Authoring

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in ntilau

ntilau

Eindhoven, Netherlands

PROJECTS

African Great Lakes legacy

08/2026 -

Florence, Italy

- From Bunyoro-Kitara Empire to modern days leveraging digital content creation.

LM9000: 65 MW aeroderivative gas turbine

Baker Hughes, a GE Company

06/2018 - 12/2018

Florence, Italy

- Contributed to conceptual instrumentation design and test rigs erection, data acquisition architecture and control system design by technical and HAZOP reviews.

Bearings loading test rig

Baker Hughes, a GE Company

05/2018 - 12/2018

Florence, Italy

- Conceptual design of a novel bearings loading test rig aimed at validating bearings NTIs.
- Investigated on modeling methods of EM coupling between power (variable speed drive system) and signals cabling systems to predict potential interference issues.

Gas Turbine embedded SAW sensors

Baker Hughes, a GE Company

02/2018 - 12/2018

Florence, Italy

- Co-advised MEng thesis student with University of Florence to investigate into SAW sensors high frequency numerical modeling for accuracy control of temperature measurements.

Cross'd

01/2018 - 11/2018

Florence, Italy

- Social networking app PoC based on Bluetooth Low Energy ranging. Made with Apache Cordova and an AWS EC2 instance

BLE Asset Tracking

07/2017 - 09/2017

Florence, Italy

- Sub 5\$/Unit PoC of BLE tags (TI's CC2640) with firmware optimized for long-battery life, aimed at asset tracking and inventory management.

GE Aviation's Catalyst

Baker Hughes, a GE Company

06/2017 - 12/2018

Prague, Czech Republic

- Led special instrumentation design and global technical reviews of FETT with design and testing teams (Czech Rep., Poland, US, Canada)
- Contributed to >4 M\$ cross-P&L project and customer satisfaction.

GE Power slip-ring digital telemetry

Baker Hughes, a GE Company

04/2017 - 05/2017

Florence, Italy

- Ran Ansys HFSS simulations to optimize signal integrity over slip-ring connections (power supply and ethernet transmission lines)
- Achieved shrinking of the rings spacing leading to manufacturing costs reduction

EDUCATION

Doctor of Philosophy - PhD, Electronics Engineering

Università degli Studi di Firenze

01/2011 - 02/2014

Florence, Italy

School of "RF, Microwaves and Electromagnetics". Computational Methods focus. With grant (Jan 2011 - Jun 2013).

Master's Degree, Electronics Engineering

Università degli Studi di Firenze

10/2007 - 04/2010

Florence, Italy

Erasmus Placement/LLP framework thesis semester at the Lehrstuhl für Theoretische Elektrotechnik, Universität des Saarlandes. With grant (Sep 2009 - Mar 2010).

Bachelor's degree, Electronics Engineering

Università degli Studi di Firenze

01/2003 - 04/2008

Florence, Italy

Thesis project at the Laboratorio di Tecnologia per i Beni Culturali, Università degli Studi di Firenze.

High School Diploma

École Autonome Belge à Bujumbura

09/1996 - 06/2002

Bujumbura, Burundi

AWARDS



EMEA TAM of the Year | Ansys (01/2021)

Award for customer and industry focus



Learn and adapt to win | BHGE (07/2018)

Patent impact award for the disclosure of "Module with sensors arrangement"



Customers determine our success | BHGE (03/2017)

Bronze award for the GE Aviation's Catalyst FETT advanced test instrumentation consulting



Customers determine our success | GE (10/2016)

Bronze award for the successful execution of the vibration monitoring system ZADCO customer structural vibrations acceptance tests



Learn and adapt to win | GE (03/2016)

Bronze award for the co-generation plant cost savings on the SNC1-2 steam turbine performance validation

- 4 weeks-FTE cross-P&L billing and customer satisfaction

Wet gas compressors at KLAB

GE Oil & Gas

📅 02/2016 - 06/2016

📍 Haugesund, Norway

- Designed and procured validation equipment for ATEX zone 1 test bench.
- Developed Ex-d antenna and low noise amplifier system for GPS synchronization on NI's PXI systems.
- Managed IEC 60079-11 simple apparatus compliance on load cells upon using extremely low voltage (1.5 V) Ex-ia Wheatstone bridge conditioners.

GE-6G PowerGen Module Structural Vibrations Monitoring System

GE Oil & Gas

📅 11/2015 - 03/2016

📍 Massa, Italy

- Developed ATEX-compliant distributed acquisition system for more than 150 vibration measurement points, with 40 kHz analog bandwidth, WiFi5 backhaul to storage and real-time monitoring, with GPS or daisy-chain clock distribution synchronization
- Hands-on design of 3D assemblies and BOM for timely delivery and cost optimization of certified equipment in less than 4 months (vs typ. 8 months).
- Introduced and consolidated with third party metrology a software based velocity and displacement real-time computation from accelerometers, achieving direct \$150k savings, and subsequent lab processes OPEX benefits.
- Customer value delivered as onsite Turbomachinery Modules structural commissioning with patented solution

SNC1-2 steam turbine performance evaluation in 6.2 MW cogeneration plant

GE Oil & Gas

📅 09/2015 - 04/2016

📍 Florence, Italy

- Defined novel pressure hookups method at lower cost and better performance to achieve accuracy controlled inter-stage enthalpy measurements according to ASME PTC 19.11.
- Programmed a CAN to Modbus TCP embedded module to enable static torque measurements on a MagCanica telemetry.

NovalT16 gas turbine, bearings and cooling component tests

GE Oil & Gas

📅 04/2014 - 10/2015

📍 Florence, Italy

- Design and validation of smartbox (field datalogger) for analog signals.
- Defined EMC-aware measurement loops design to achieve noise immunity and enhanced measurement accuracies, rigorously applying EN 61000 guidelines (CEN-ELEC25).

Power density centrifugal compressor

GE Oil & Gas

📅 03/2014 - 05/2016

📍 Florence, Italy

- Led the commissioning Fieldbus Foundation use in testing lab for more than 240 pressure transducers (0.1% uncertainty) upon identifying the proper Firmware configuration which enabled for massive and accurate pressure measurements.
- Managed capacitive vibration probes ATEX-compliance.
- Developed electrostatic numerical model to predict capacitance variation in various surroundings and target shape conditions.
- Contributed to DM37/08 compliance of the test rig contract specification, including power and signaling cabling selection and proper installation.

GE Aviation's GE9X High Pressure Compressor

GE Oil & Gas

📅 01/2014 - 04/2014

📍 Massa, Italy



Learn and adapt to win | GE (10/2015)

Bronze award for the introduction and validation of an effective, cost-saving, structural vibrations digital integration algorithm aimed at accurately deriving speed & displacements measurements from ATEX-compliant accelerometers



External focus | GE (06/2014)

Bronze award for the RCA activities on Cessao Onerosa, ADRE monitoring system EMI noise mitigation and customer assurance



Giorgio Barzilai Prize | SIEm (09/2012)

Best paper award for young researchers, for the work "Model order reduction in Finite Element analysis of phased array antennas" presented at the XIX RiNEM



Ph.D. studies grant | UniFi (12/2010)

Full tuition fees and allowance for research on Electromagnetic Compatibility modeling techniques in variable speed drive systems - with GE/Nuovo Pignone



Dean's listing | UniFi (04/2010)

Commissione di Laurea Specialistica in Ingegneria Elettronica presieduta dal Prof. Piero Tortoli (Prot. 1051 n. class. III/9.1.2 del 24 Maggio 2010)



LLP/Erasmus Placement grant | UniFi (10/2009)

Thesis work semester mobility support at Universität des Saarlandes

CERTIFICATIONS



Professional Engineer,
ID:ES2010277049000038

Università degli Studi di Firenze

COURSES

Command of the Message

Ansyz

📅 06/2020

📍 Remote

JAWS: Just Another Way of Selling

Ansyz

📅 02/2020

📍 Paris, France

HBM Dynamic Load Cells

GE Oil & Gas

📅 05/2016

📍 Bologna, Italy

GE Crotonville: Delivering Customer Impact

GE Oil & Gas

📅 03/2016

📍 Munich, Germany

GE Control Valves — Advanced

GE Oil & Gas

📅 09/2015

📍 Florence, Italy

- High density data acquisition system design for GE Aviation's compressor test in Massa.
- Performed mitigation on EMC noise due to 130 kW variable speed ventilation system

BCL306 and BCL317 centrifugal compressors rotordynamics

GE Oil & Gas

06/2013 - 03/2014 Florence, Italy

- Performed an Ex-p execution of stability's magnetic exciter, contributed to the resolution of customer's opened nonconformity on rotordynamic vibrations upon conducting an electromagnetic interference on Bently Nevada's 3500 RCA.

Finite Element Software (FES)

Università degli Studi di Firenze

06/2011 - 02/2014 Florence, Italy

- FES is an open source framework for the development of "ad-hoc" simulation software aimed at a rapid implementation of novel numerical methods based on the Galerkin method. Core formulations verified reusing meshes from commercial packages (Comsol, HFSS)

Phased array antennas model order reduction (MEng thesis)

Universität des Saarlandes

10/2009 - 02/2010 Saarbrücken, Germany

- Developed in Matlab efficient near-to-far-fields formulations for fast electromagnetic radiation computation in large finite elements problems exploiting model order reduction with spectral interpolation for beam steering and pattern angles
- Modeled a large patch antenna array, assessing accuracy and performances through mesh/model export from Ansoft HFSS v11

RS-232 modem in ISM 868 MHz band (BEng thesis)

University of Florence

01/2007 - 04/2007 Florence, Italy

- Designed and prototyped an 868 MHz wireless link for a ground penetration radar remote control over RS-232 protocol
- C-coded the PIC microcontroller FW and optimized in Assembly to guarantee the full-duplex operation over half-duplex TI's CC1100 radio.
- Evaluated the performance of the wireless link in open field

CE Marking Overview

GE Oil & Gas

11/2014 Florence, Italy

Safety Integrity Levels

GE Oil & Gas

10/2014 Florence, Italy

Electrification

GE Oil & Gas

10/2014 Florence, Italy

Finite Element Software (FES)

GE Oil & Gas

10/2013 Florence, Italy

ATEX Intermediate

GE Oil & Gas

09/2014 Florence, Italy

Total Quality Lean Six Sigma Green Belt

GE Oil & Gas

06/2014 Florence, Italy

PidXp and P&ID

GE Oil & Gas

11/2013 Florence, Italy

Intellectual Property

GE Oil & Gas

11/2013 Florence, Italy

GE Oil & Gas Product Familiarization

GE Oil & Gas

07/2013 Florence, Italy

SKILLS

Engineering

Simulations

Testing

Management

Sales

Strategy

Leadership

Customer Satisfaction

Finite Element Analysis

RF

Electronics

Semiconductors

Scripting

Matlab

C++

C

Python

LANGUAGES

Dutch

English

French

German

Italian

Kirundi

PUBLICATIONS

Patents

- L. Ntibarikure, "Module with sensors arrangement," US20180180464A1, 2018. [Online]. Available: <https://patents.google.com/patent/US20180180464A1/en>

Books

- L. Ntibarikure, *Contributions to the Art of Finite Elements in Electromagnetics*. Florence, Italy, 2014. [Online]. Available: <https://hdl.handle.net/2158/843133>

Journal Articles

- L. Ntibarikure, "Multiphysics simulations for 5g rfrics and socs," *Bits&Chips*, 2019.
- F. Barone, A. Signorini, L. Ntibarikure, T. Fiore, F. Di Pasquale, and C. J. Oton, "Fiber-optic liquid level sensing by temperature profiling with an fbg array," *Sensors*, vol. 18, no. 8, 2018. [Online]. Available: <https://www.mdpi.com/1424-8220/18/8/2422>
- L. Ntibarikure, G. Pelosi, and S. S. and, "Harmonic balance domain decomposition finite elements for nonlinear passive microwave devices analysis," *Electromagnetics*, vol. 34, no. 3-4, pp. 239–252, 2014. DOI: 10.1080/02726343.2014.877756
- L. Ntibarikure, G. Pelosi, and S. Selleri, "Efficient harmonic balance analysis of waveguide devices with nonlinear dielectrics," *IEEE Microwave and Wireless Components Letters*, vol. 22, no. 5, pp. 221–223, 2012. DOI: 10.1109/LMWC.2012.2192420

Conference Proceedings

- L. Ntibarikure, G. Pelosi, and S. Selleri, "Assessment of the performances of gmres(r) using a domain decomposition approach as a preconditioner," in *Proceedings, XX Riunione Nazionale di Elettromagnetismo (XX RiNEm)*, Padua, Italy, 2014. [Online]. Available: <https://inspirehep.net/literature/1403858>
- L. Ntibarikure, "Model order reduction in finite elements analysis of phased array antennas," in *Proceedings, XIX Riunione Nazionale di Elettromagnetismo (XIX RiNEm)*, S. d. E. SiEm, Ed., Roma, Italy, 2012.
- L. Ntibarikure, G. Pelosi, and S. Selleri, "Harmonic balance finite element analysis of third order intermodulation products in ferrite devices," in *Proceedings, XIX Riunione Nazionale di Elettromagnetismo (XIX RiNEm)*, S. d. E. SiEm, Ed., Roma, Italy, 2012.